

Rotating medium — Solution

W26 (2026) — English translation

A1

1.30 pts

Find the trajectory of motion $\vec{r}(t)$ for this particle. Express the answer in vector form through $\vec{v}_0$, $\vec{g}$, ε or express the dependences of the components of the vector $\vec{r}(t)$ in the Cartesian coordinate system through the components of the vectors $\vec{v}_0$, $\vec{g}$.

Newton's second law:

$$m\dot{\vec{v}} = -\varepsilon m\vec{v} + m\vec{g}.$$

Разобьём velocity на два слагаемых:

$$\vec{v}(t) = \vec{u}(t) + \frac{\vec{g}}{\varepsilon}.$$

Equation на $\vec{u}$:

$$\begin{aligned} \dot{\vec{u}} &= -\varepsilon\vec{u}. \\ \vec{v}(t) &= \frac{\vec{g}}{\varepsilon} + \left(\vec{v}_0 - \frac{\vec{g}}{\varepsilon}\right) e^{-\varepsilon t} \end{aligned}$$

Answer: \textbf{Answer:}

$$\vec{r}(t) = \frac{\vec{g}t}{\varepsilon} + \left(\vec{v}_0 - \frac{\vec{g}}{\varepsilon}\right) \frac{1 - e^{-\varepsilon t}}{\varepsilon}$$

A2

0.70 pts

How does the power of energy loss $P(t)$, which goes into heat, depend on time? Express your answer in terms of $\vec{v}_0$, $\vec{g}$, ε , t , m .

Let us write the law of change in the energy of the particle+medium system:

$$m(\vec{v}, \dot{\vec{v}}) = m(\vec{v}, \vec{g}) - P.$$

$$P = -m(\vec{v}, \dot{\vec{v}} - \vec{g}) = \varepsilon m v^2$$

$$\vec{v}(t) = \frac{\vec{g}}{\varepsilon} (1 - e^{-\varepsilon t}) + \vec{v}_0 e^{-\varepsilon t}$$

$$v^2 = \frac{g^2 (1 - e^{-\varepsilon t})^2}{\varepsilon^2} + v_0^2 e^{-2\varepsilon t} + \frac{2(\vec{v}_0, \vec{g})}{\varepsilon} (1 - e^{-\varepsilon t}) e^{-\varepsilon t}$$

Answer: \textbf{Answer:}

$$P = \varepsilon m \left(\frac{g^2 (1 - e^{-\varepsilon t})^2}{\varepsilon^2} + v_0^2 e^{-2\varepsilon t} + \frac{2(\vec{v}_0, \vec{g})}{\varepsilon} (1 - e^{-\varepsilon t}) e^{-\varepsilon t} \right)$$

A3

2.00 pts

Find the loss power P , which goes into heat, a long time after the power $t \gg 1/\varepsilon$ is turned on. Express your answer using F_0 , ε , m , ω .

During time $t \gg 1/\varepsilon$, motion is established, and from the isotropy of the system it is

clear that motion will occur in a circle with an angular velocity ω and a constant modulus velocity u . In this case, the angle between speed and force is constant and equal to α . Let's write down Newton's second law in projection onto the direction of velocity and the normal to it.

$$\begin{cases} 0 = F_0 \cos \alpha - \varepsilon m u \\ m u \omega = F_0 \sin \alpha \end{cases}$$

$$\tan \alpha = \frac{\omega}{\varepsilon}$$

$$u = \frac{F_0 \cos \alpha}{\varepsilon m} = \frac{F_0}{\varepsilon m} \frac{\varepsilon}{\sqrt{\varepsilon^2 + \omega^2}} = \frac{F_0}{m \sqrt{\varepsilon^2 + \omega^2}}$$

Закон изменения энергии for системы:

$$P = F_0 u \cos \alpha = \frac{F_0^2 \varepsilon}{m (\varepsilon^2 + \omega^2)}.$$

Answer: \textbf{Answer:}

$$P = \frac{F_0^2 \varepsilon}{m (\varepsilon^2 + \omega^2)}$$

B1

0.50 pts

Find the dependence $z(t)$. Express your answer in terms of r_0 , ε , g .

Projection of Newton's second law in projection onto the z axis:

$$m \ddot{z} = -mg - \varepsilon m \dot{z}.$$

$$\ddot{z} = -g - \varepsilon \dot{z}$$

We solve similarly to the previous part of the problem.

Answer: \textbf{Answer:}

$$z(t) = -\frac{gt}{\varepsilon} + \frac{g(1 - e^{-\varepsilon t})}{\varepsilon^2}$$

B2

1.00 pts

Express $\ddot{x}$ and $\ddot{y}$ in terms of x , y , $\dot{x}$, $\dot{y}$, ε and ω .

Newton's second law:

$$m \ddot{\vec{r}} = m \vec{g} + \varepsilon m \left([\vec{\omega}, \vec{r}] - \dot{\vec{r}} \right).$$

В проекции на оси x and y :

Answer: \textbf{Answer:}

$$\begin{cases} \ddot{x} = \varepsilon (-\omega y - \dot{x}) \\ \ddot{y} = \varepsilon (\omega x - \dot{y}) \end{cases}$$

B3

1.00 pts

Using the solution (x_1, y_1) , obtain a system of equations from which the possible values of the constants a and Ω can be determined. The answer may also include ε , ω .

First method

Note that it is enough to consider one of the solutions, since the other is obtained by rotating the system relative to the z axis, therefore, from symmetry, the dependence coefficients will not change. Let's substitute the first solution option into the answer from

the previous paragraph.

$$\begin{pmatrix} x \\ y \end{pmatrix} = Ae^{at} \begin{pmatrix} \cos(\Omega t) \\ \sin(\Omega t) \end{pmatrix}$$

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = Ae^{at} \begin{pmatrix} a \cos(\Omega t) - \Omega \sin(\Omega t) \\ a \sin(\Omega t) + \Omega \cos(\Omega t) \end{pmatrix}$$

$$\begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = Ae^{at} \begin{pmatrix} (a^2 - \Omega^2) \cos(\Omega t) - 2a\Omega \sin(\Omega t) \\ (a^2 - \Omega^2) \sin(\Omega t) + 2a\Omega \cos(\Omega t) \end{pmatrix}$$

Substituting в answer прошлого parta:

$$\begin{pmatrix} (a^2 - \Omega^2) \cos(\Omega t) - 2a\Omega \sin(\Omega t) \\ (a^2 - \Omega^2) \sin(\Omega t) + 2a\Omega \cos(\Omega t) \end{pmatrix} = \begin{pmatrix} -\varepsilon\omega \sin(\Omega t) \\ \varepsilon\omega \cos(\Omega t) \end{pmatrix} - \begin{pmatrix} a\varepsilon \cos(\Omega t) - \varepsilon\Omega \sin(\Omega t) \\ a\varepsilon \sin(\Omega t) + \varepsilon\Omega \cos(\Omega t) \end{pmatrix}$$

For/Атравниваем coefficients перед синусами and косинусами we get систему equation на a and Ω :

$$\begin{cases} a^2 - \Omega^2 = -a\varepsilon \\ 2a\Omega = \varepsilon\omega - \varepsilon\Omega \end{cases}$$

Second method

Let us consider plane motion in polar coordinates (r, φ) .

$$\begin{aligned} \dot{\vec{r}} &= \dot{r}\vec{e}_r + \dot{\varphi}r\vec{e}_\varphi \\ \ddot{\vec{r}} &= (\ddot{r} - \dot{\varphi}^2 r)\vec{e}_r + (\ddot{\varphi}r + 2\dot{r}\dot{\varphi})\vec{e}_\varphi \end{aligned}$$

Let us write Newton's second law в проекции на $\vec{e}_r$ and $\vec{e}_\varphi$:

$$\begin{cases} \ddot{r} - \dot{\varphi}^2 r = -\varepsilon\dot{r} \\ \ddot{\varphi}r + 2\dot{r}\dot{\varphi} = \varepsilon(\omega r - \dot{\varphi}r) \end{cases}$$

Substituting $r = Ae^{at}$ and $\varphi = \Omega t$:

$$\begin{cases} a^2 - \Omega^2 = -\varepsilon a \\ 2a\Omega = \varepsilon\omega - \varepsilon\Omega \end{cases}$$

We obtained a similar system of equations the first method.

Answer: \textbf{Answer:}

$$\begin{cases} a^2 - \Omega^2 = -\varepsilon a \\ 2a\Omega = \varepsilon\omega - \varepsilon\Omega \end{cases}$$

B4

1.00 pts

Solve the resulting system of equations and find suitable pairs of values (a_1, Ω_1) and (a_2, Ω_2) if $a_2 > a_1$. Express your answer using ε, ω .

To solve, we transform the equations:

$$\begin{cases} \Omega^2 = \left(a + \frac{\varepsilon}{2}\right)^2 - \frac{\varepsilon^2}{4} \\ \Omega = \frac{\varepsilon\omega}{2\left(a + \frac{\varepsilon}{2}\right)} \end{cases}$$

Denote $\gamma = \left(a + \frac{\varepsilon}{2}\right)^2$.

$$\begin{aligned} \frac{\varepsilon^2\omega^2}{4\gamma} &= \Omega^2 = \gamma - \frac{\varepsilon^2}{4} \\ \gamma^2 - \frac{\varepsilon^2}{4}\gamma &= \frac{\varepsilon^2\omega^2}{4} = \left(\gamma - \frac{\varepsilon^2}{8}\right)^2 - \frac{\varepsilon^4}{64} \\ \gamma &= \frac{\varepsilon^2}{8} \pm \sqrt{\frac{\varepsilon^2\omega^2}{4} + \frac{\varepsilon^4}{64}} \end{aligned}$$

Note that $\gamma > 0$, therefore подходит только корень со знаком +.

$$a + \frac{\varepsilon}{2} = \pm \sqrt{\frac{\varepsilon^2}{8} + \sqrt{\frac{\varepsilon^2 \omega^2}{4} + \frac{\varepsilon^4}{64}}}$$

$$a = \varepsilon \left(-\frac{1}{2} \pm \sqrt{\frac{1}{8} + \sqrt{\frac{\omega^2}{4\varepsilon^2} + \frac{1}{64}}} \right)$$

$$\Omega = \pm \frac{\omega}{\sqrt{\frac{1}{2} + \sqrt{\frac{4\omega^2}{\varepsilon^2} + \frac{1}{4}}}}$$

Answer: \textbf{Answer:}

$$a_1 = \varepsilon \left(-\frac{1}{2} - \sqrt{\frac{1}{8} + \sqrt{\frac{\omega^2}{4\varepsilon^2} + \frac{1}{64}}} \right)$$

$$a_2 = \varepsilon \left(-\frac{1}{2} + \sqrt{\frac{1}{8} + \sqrt{\frac{\omega^2}{4\varepsilon^2} + \frac{1}{64}}} \right)$$

$$\Omega_1 = -\frac{\omega}{\sqrt{\frac{1}{2} + \sqrt{\frac{4\omega^2}{\varepsilon^2} + \frac{1}{4}}}}$$

$$\Omega_2 = \frac{\omega}{\sqrt{\frac{1}{2} + \sqrt{\frac{4\omega^2}{\varepsilon^2} + \frac{1}{4}}}}$$

B5

0.50 pts

Find approximate values of these quantities in the cases $\varepsilon \ll \omega$ and $\varepsilon \gg \omega$.

Answer: \textbf{Answer:} For $\varepsilon \ll \omega$:

$$a_1 = -\sqrt{\frac{\omega\varepsilon}{2}} = \Omega_1$$

$$a_2 = \sqrt{\frac{\omega\varepsilon}{2}} = \Omega_2$$

For $\varepsilon \gg \omega$:

$$a_1 = -\varepsilon$$

$$a_2 = 0$$

$$\Omega_1 = -\omega$$

$$\Omega_2 = \omega$$

B6

1.50 pts

An arbitrary solution to the system of equations has the form

$$(x(t), y(t)) = A\vec{r}_{11} + B\vec{r}_{12} + C\vec{r}_{21} + D\vec{r}_{22}.$$

Using начальные условия, for/атведенные во введении к части, выtimesandme A , B , C and D via/through r_0 , a_1 , a_2 , Ω_1 and Ω_2 .

Initial conditions: $x(0) = 0$, $y(0) = r_0$, $\dot{x}(0) = 0$ and $\dot{y}(0) = 0$.

$$\begin{cases} A + B = r_0 \\ C + D = 0 \\ a_1 A + a_2 B - \Omega_1 C - \Omega_2 D = 0 \\ a_1 C + a_2 D + \Omega_1 A + \Omega_2 B = 0 \end{cases}$$

$$D = -C$$

$$\begin{cases} A + B = r_0 \\ a_1 A + a_2 B = C (\Omega_1 - \Omega_2) \\ \Omega_1 A + \Omega_2 B = C (a_2 - a_1) \end{cases}$$

$$\frac{a_1 A + a_2 B}{\Omega_1 A + \Omega_2 B} = \frac{\Omega_1 - \Omega_2}{a_2 - a_1}$$

$$A (a_1 (a_2 - a_1) + \Omega_1 (\Omega_2 - \Omega_1)) = B (a_2 (a_1 - a_2) + \Omega_2 (\Omega_1 - \Omega_2))$$

$$r_0 = A \frac{a_2 (a_1 - a_2) + \Omega_2 (\Omega_1 - \Omega_2) + a_1 (a_2 - a_1) + \Omega_1 (\Omega_2 - \Omega_1)}{a_2 (a_1 - a_2) + \Omega_2 (\Omega_1 - \Omega_2)}$$

$$r_0 = A \frac{(a_2 - a_1)^2 + (\Omega_2 - \Omega_1)^2}{a_2 (a_2 - a_1) + \Omega_2 (\Omega_2 - \Omega_1)}$$

$$A = r_0 \frac{a_2 (a_2 - a_1) + \Omega_2 (\Omega_2 - \Omega_1)}{(a_2 - a_1)^2 + (\Omega_2 - \Omega_1)^2}$$

$$B = -r_0 \frac{a_1 (a_2 - a_1) + \Omega_1 (\Omega_2 - \Omega_1)}{(a_2 - a_1)^2 + (\Omega_2 - \Omega_1)^2}$$

$$C = \frac{a_1 A + a_2 B}{\Omega_1 - \Omega_2} = r_0 \frac{a_2 \Omega_1 - a_1 \Omega_2}{(a_2 - a_1)^2 + (\Omega_2 - \Omega_1)^2}$$

$$D = r_0 \frac{a_1 \Omega_2 - a_2 \Omega_1}{(a_2 - a_1)^2 + (\Omega_2 - \Omega_1)^2}$$

Answer: \textbf{Answer:}

$$A = r_0 \frac{a_2 (a_2 - a_1) + \Omega_2 (\Omega_2 - \Omega_1)}{(a_2 - a_1)^2 + (\Omega_2 - \Omega_1)^2}$$

$$B = -r_0 \frac{a_1 (a_2 - a_1) + \Omega_1 (\Omega_2 - \Omega_1)}{(a_2 - a_1)^2 + (\Omega_2 - \Omega_1)^2}$$

$$C = r_0 \frac{a_2 \Omega_1 - a_1 \Omega_2}{(a_2 - a_1)^2 + (\Omega_2 - \Omega_1)^2}$$

$$D = r_0 \frac{a_1 \Omega_2 - a_2 \Omega_1}{(a_2 - a_1)^2 + (\Omega_2 - \Omega_1)^2}$$

B7

0.50 pts

Draw qualitatively the trajectory of the particle in the xy plane for cases $\varepsilon \ll \omega$ and $\varepsilon \gg \omega$.

For $\varepsilon \ll \omega$ we get:

$$x(t) = r_0 \cos \Omega_2 t \cosh \Omega_2 t$$

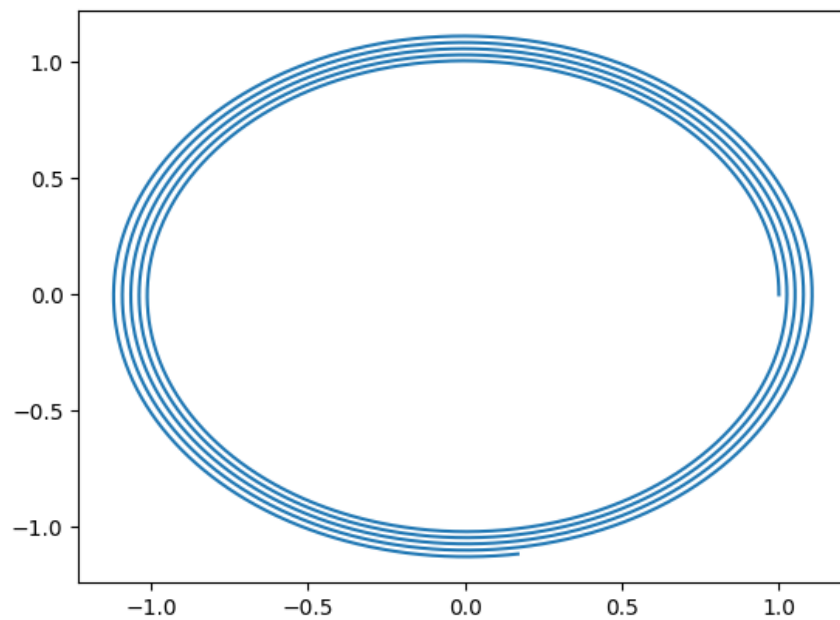
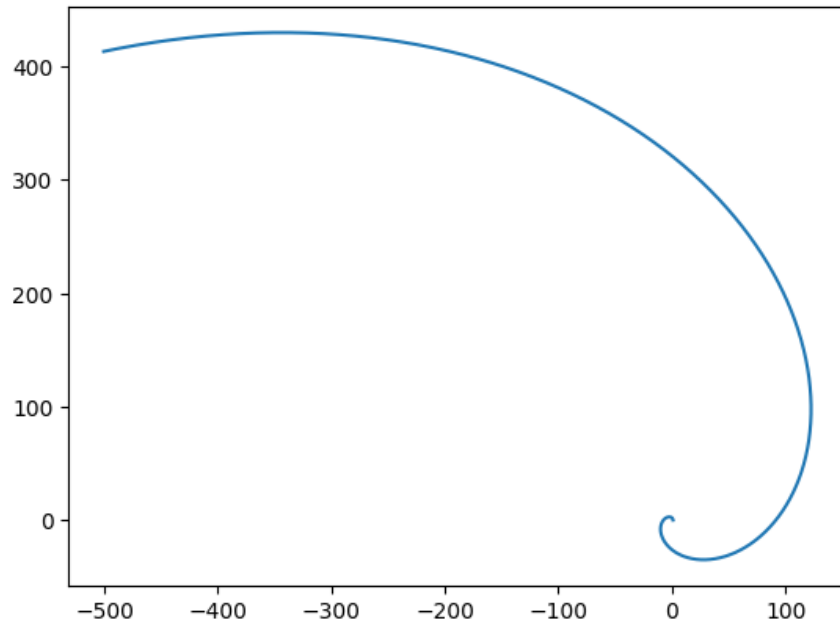
$$y(t) = r_0 \sin \Omega_2 t \sinh \Omega_2 t$$

For/At пренебрежении затухающей экспонентой получается логарифмическая спираль. Angle между radius-vectorом and velocityю via/through большое time после начала движения equals 45° .

At $\varepsilon \gg \omega$ the particle moves along a circle of radius r_0 with a slowly increasing radius:

$$x(t) \approx r_0 \cos \omega t$$

$$y(t) \approx r_0 \sin \omega t$$



B8

1.00 pts

Find the energy loss power P , which goes into heat, at the moment when the particle is at a point with coordinates $\vec{r} = (x, y, z)$ and moves with a speed $\vec{v} = (v_x, v_y, v_z)$.

Let us write down the law of change in the energy of the system:

$$m \left(\vec{v}, \dot{\vec{v}} \right) = m \left(\vec{g}, \vec{v} \right) - P + W,$$

where W - power внешних сил, действующая на среду. Let us find W . Чтобы medium вращалась в постоянной угловой velocity, на каждый её учасcurrent должна действовать force с нулевой проекцией на направление скорости, therefore на область, в которой находится частица, действует force $-\vec{F}$ со стороны частицы and внешняя force $\vec{F}$.

$$W = \left(\vec{F}, [\vec{\omega}, \vec{r}] \right)$$

$$\left(\vec{v}, m\vec{g} + \vec{F} \right) = m \left(\vec{g}, \vec{v} \right) - P + \left(\vec{F}, [\vec{\omega}, \vec{r}] \right)$$

$$P = \left(\vec{F}, [\vec{\omega}, \vec{r}] - \vec{v} \right) = \varepsilon m v_{\text{отн}}^2 = \varepsilon m \left(v_z^2 + (v_x + \omega y)^2 + (v_y - \omega x)^2 \right)$$

Answer: \textbf{Answer:

$$P = \left(\vec{F}, [\vec{\omega}, \vec{r}] - \vec{v} \right) = \varepsilon m v_{\text{отн}}^2 = \varepsilon m \left(v_z^2 + (v_x + \omega y)^2 + (v_y - \omega x)^2 \right)$$